

1.

Use ssh to connect from your Linux system to a remote server (you can use a free-tier cloud VM or a local VM). After connecting, create a directory named 'music_downloads' in your home folder on the remote server.

Ans : **Connect to Remote Server and Create Directory**

```
ssh username@server_ip
```

Create the directory:

```
mkdir ~/music_downloads
```

2.

Transfer a file named 'playlist.txt' from your local machine to the remote server's 'music_downloads' directory using the scp command.

Hint: Check the syntax for scp to specify both source and destination paths.

Ans : **Copy playlist.txt Using SCP**

```
scp playlist.txt username@server_ip:~/music_downloads/
```

3.

Set up passwordless SSH login from your local Linux system to the remote server by generating an SSH key pair and copying the public key to the remote server.

Constraint: Do not use a passphrase for the key during this setup.

Ans : **Set Up Passwordless SSH Login**

Generate SSH key (no passphrase):

```
ssh-keygen -t rsa
```

Press **Enter** when asked for a passphrase.

Copy the key:

```
ssh-copy-id username@server_ip
```

Now connect without a password:

```
ssh username@server_ip
```

4.

Simulate a Zomato-style admin task: Write a bash script on your local machine that uses ssh to remotely restart the 'nginx' service on the remote server and then checks its status. Save your script as 'remote_nginx_check.sh'.

Ans : **remote_nginx_check.sh**

```
#!/bin/bash
```

```
ssh username@server_ip "sudo systemctl restart nginx && sudo systemctl status nginx"
```

Make it executable:

```
chmod +x remote_nginx_check.sh
```

Run:

```
./remote_nginx_check.sh
```

5.

Find and fix the error in this ssh command: ssh

user@192.168.1.100:/home/user/music_downloads

Hint: Review the correct syntax for connecting to a remote server with ssh.

Ans : Fix the SSH Command

Incorrect

```
ssh user@192.168.1.100:/home/user/music_downloads
```

Correct:

```
ssh user@192.168.1.100
```

After connecting:

```
cd /home/user/music_downloads
```